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ABSTRACT

This project was assigned to me by Ford Otosan which is an automotive company that operates mainly in Turkey. I was working in ADAS team which works for developing autonomous vehicles. I was working on a 3D vision detection algorithm called BEVFusion in order to use it in autonomous vehicles. I made a literature research on vision detection algorithms and made an inference with BEVFusion by using the nuScenes dataset.

Unfortunately, I couldn't complete CUDA & TensorRT accelerated BEVFusion due to hardware compatibility problems, and I didn't try it on an Nvidia Orin Drive is used to do inference on real autonomous cars. The remaining parts of the project can be completed by a future intern. Even though the internship was short, it was a good experience to get familiar with vision recognition and deep learning.

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1. INTRODUCTION

In this paper, I will talk about BEVFusion Deployment Project that I did at Ford Otosan during my summer internship. BEVFusion is an important 3D vision detection algorithm. It combines inputs from cameras and LIDAR to create 3D bounding boxes around objects. Its hybrid detector approach enables it to guess bounding boxes in a high accuracy and makes it a useful algorithm for autonomous vehicles. In the following pages on this paper, I will talk more about the BEVFusion and the project. Firstly, there will be some background information about Ford Otosan. Secondly, it will continue with description and analysis of the internship such as the purpose of the project, things that I have done and the results. Thirdly, I will share my observations and analysis on the internship. For instance, I will talk about, is there any correlation between internship experience and SU classes and finally my paper will be concluded with a conclusion which will summarize the important points of the paper and discuss the results of the project.

2. COMPANY BACKGROUND

Ford Otosan is a publicly traded company which was founded by Koç Holding and Ford Motor Company. It mainly operates in the automotive sector and produces trucks and transit vehicles. For instance, Ford Transit is “the longest running model in Ford Europe’s portfolio, produced for more than 50 years” (*Vehicle Production*, n.d.)." It also has RD projects on engines and vehicles. It currently operates in six locations: Gölcük and Yeniköy plants in Kocaeli, İnönü plant in Eskişehir, Craiova plant in Romania, Sancaktepe R&D Center and spare parts warehouse in Istanbul (Wikipedia contributors, 2025). The company employs more than 20,000 people and had a production capacity of over 700,000 vehicles,

400,000 engines, and 140,000 powertrains by 2022 (Wikipedia contributors, 2025). I was in the AR-Ge department specifically at ADAS (Advanced driver-assistance systems) team.

Their main aim was developing autonomous and semi-autonomous vehicles. I was in the perception team which was responsible of sensors and object detection.

3. PROJECT DESCRIPTION and ANALYSIS

3.1 PROJECT PURPOSE

Ford Otosan tries to develop autonomous (L4 level) and semi-autonomous(L2-level) vehicles. In order to do this, they need efficient and accurate vision detection algorithms to capture obstacles early on and move the car accordingly to prevent accidents. BEVFusion which is the algorithm that I worked on this project is developed for detecting objects and generating 3D boxes around them. Company selected BEVFusion algorithm because of its superior accuracy thanks to its architecture that uses both camera and LIDAR input.

3.2 PROJECT DESCRIPTION

I aimed to work on four main aspect. Firstly, I made research about both 2D and 3D detectors in order to gain insight at the field. Secondly, I used BEVFusion repo for inference and visualization purposes in order to test how it works. In order to do this first I needed to install necessary dependencies and fix the code because the owner of the repo had broken the BEVFusion code while trying to add Radar detectors along with cameras and LIDAR. Thirdly, I worked on Accelerated Inference with CUDA & TensorRT. I aimed to do inference with this accelerated version and compare the results with the previous BEVFusion model in order to observe speed accuracy trade of between them. Lastly, if I had time, I would work on setting up CUDA & TensorRT accelerated BEVFusion on Nvidia Orin Drive. Even though it

is easier to execute vision recognition algorithms in computer, for autonomous vehicles you need to do inference on more limited hardware. This is why I also wanted to do inference on Nvidia Orin because these drivers are loaded on the autonomous car for vision recognition purposes.

3.3 TECHNICAL ASPECTS

First of all, let's talk about how BEVFusion works. As I said BEVFusion uses both camera and LIDAR input to generate bounding boxes. It does this by unifying multi-modal features (from cameras and LiDAR) in a shared Bird-Eye-View (BEV) representation space. This unified space preserves both geometric (from LiDAR) and semantic (from cameras) information. Because of this reason it has a higher accuracy than other 3D detection algorithms. In order to do inference on BEVFusion, first github repo was copied. Then Nvidia toolkit 11.1 was downloaded. After that the virtual environment was created and necessary libraries were downloaded at specific versions in order to prevent library conflict. Then by using setup.py script at the repo I set up the BEVFusion. Then v-1.0 mini from nuScenes dataset was downloaded. After that .pkl files were created for the final step before the inference. Then I did inference. Firstly, I executed the detection command. Secondly, I executed the segmentation command. Lastly, I visualize our inference to better understand our results.

For doing Accelerated Inference with CUDA & TensorRT, the github repo was again copied. I downloaded same data set. Then, the previous virtual environment was copied and was used by adding some extra dependencies and installing tensorRT. After that, I run to execute Cuda accelerated inference but due to hardware problems I got an error. The GPU I used weren't compatible to run this model.

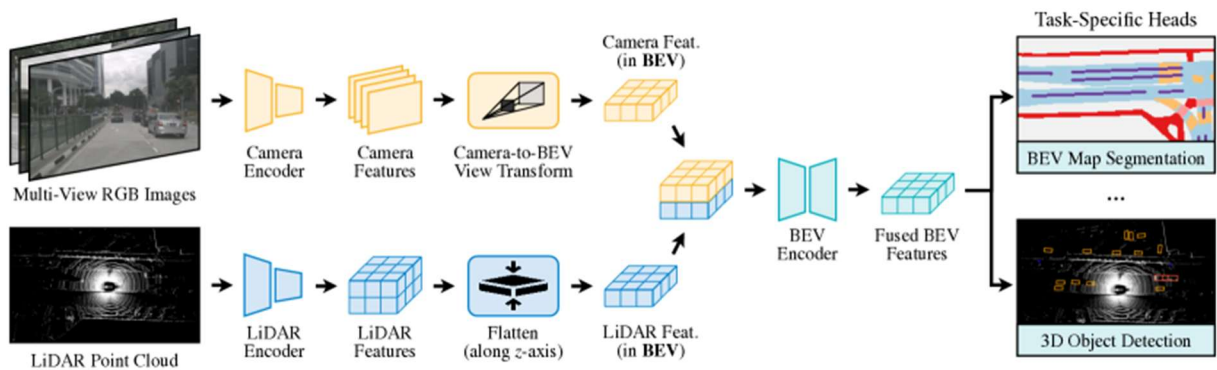


Figure 1: How BEVFusion works

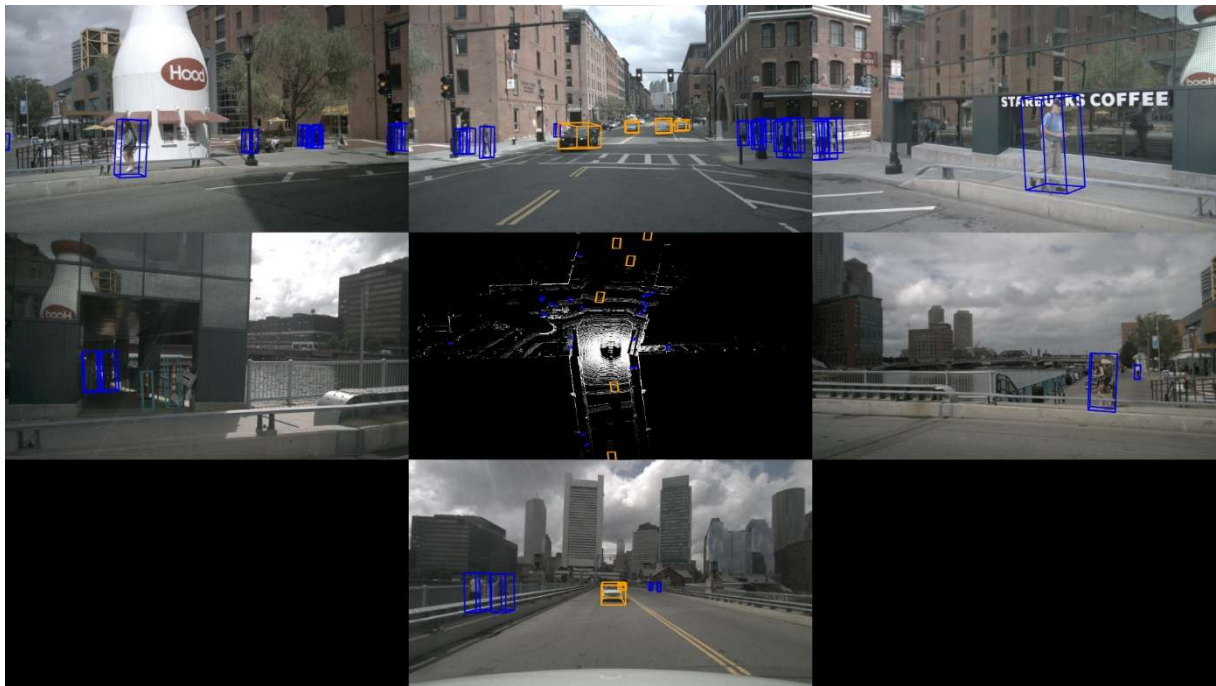


Figure 2: Visualisation of the BEVFusion Inference

3.4 METHODOLOGY and TOOLS

I used VS Code for both connecting remote computers for inference and editing code. Python was the main language of the scripts of the BEVFusion. I set up and used Nvidia Toolkit, TensorRT for inference. I installed a software called Netbird to connect my home computer via ssh because my laptop didn't have an Nvidia GPU.

3.5 PROJECT RESULTS

I made an extensive literature research on both 2D and 3D vision detection algorithms and delivered both documents and slides to company. I successfully made inference on BEVFusion by using nuScenes mini data set and create a visualization of it. I delivered a document on how to run inference code, which errors you can encounter and how to fix those for both BEVFusion Inference and Visualization and Accelerated Inference with CUDA & TensorRT. So, at the end I successfully finished researching about vision detection algorithms and BEVFusion Inference, I partially finished Accelerated BEVFusion Inference due to hardware problems and I didn't have time to set up BEVFusion on Nvidia Orin Drive. The uncompleted parts of the project will be completed by another employee or a future intern.



Figure 3: NVIDIA DRIVE AGX Orin Kit

3.6 LITERATURE REVIEW

During the project I researched both 2D and 3D detectors. For 2D detectors, I used Geeks for Geeks for understanding two stage detectors and difference between them (GeeksforGeeks, 2023). Then, I used a medium article to compare one stage detectors with two stage detectors (Hussain, 2024). For 3D detectors, I used Lee and other's "A Survey for 3D Object Detection Algorithms from Images" (Lee et al., 2022) in order understand different types of 3D detector families and difference between them. I also used Chen and other's "2D and 3D object detection algorithms from images" (Chen et al., 2023) in order to learn about both 2D and 3D detectors. Moreover, I also used the journals of the developer teams who invented the algorithms itself. For learning about Monocular Image base algorithms I used Chen's "Monocular 3D Object Detection for Autonomous Driving" (X. Chen et al., 2016), Zhang's "Objects are Different: Flexible Monocular 3D Object Detection" (Brazil & Liu, 2019), Brazil and Liu's "Monocular 3D Region Proposal Network for Object Detection" (Brazil & Liu, 2019), Liu's "Learning Auxiliary Monocular Contexts Helps Monocular 3D Object Detection" (Liu et al., 2022). For learning about Stereo Image based algorithms, I used Li's "Stereo R-CNN based 3D Object Detection for Autonomous Driving" (Li et al., 2019) and Y. Chen's "DSGN: Deep Stereo Geometry Network for 3D Object Detection" (Y. Chen et al., 2020). For learning about Multi-camera Image based algorithms, I used Z. Li's "BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers" (Z. Li et al., 2022), Zhijian's "BEVFusion: Multi-Task Multi-Sensor Fusion with Unified Bird's-Eye View Representation" (Zhijian et al., 2022), Y. Liu's "PETR: Position Embedding Transformation for Multi-View 3D Object Detection" (Y. Liu et al., 2022). For learning about Fusion based image detection algorithms, I used Sindagi's "MVX-Net: Multimodal VoxelNet for 3D Object Detection" (Sindagi et al., 2019), Vora's

“PointPainting: Sequential Fusion for 3D Object Detection” (Vora et al., 2020), Bai’s “TransFusion: Robust LiDAR-Camera Fusion for 3D Object Detection with Transformers” (Bai et al., 2022). For learning about Keypoint based 3D image recognition algorithms, I used Wang’s “CenterNet3D: An Anchor Free Object Detector for Point Cloud” (Wang et al., 2021).

4. INTERNSHIP EXPERIENCE

4.1 RELATION to UNDERGRADUATE CURRICULLUM

This internship was mainly focused on vision recognition algorithms and deep learning. Because of this reason CS 415 (Introduction to deep learning), CS 419 (Digital Image and Video Analysis), EE 417 (Computer Vision) would be incredibly useful but unfortunately, I didn’t get those courses before starting my internship. After the internship, I gained interest in these topics, and started to think about enrolling these courses. I had enrolled DSA 210 and CS 412 in my past semesters and I used some of the basic information that I learned there such as training/test set, max pooling, neural networks.

4.2 DIFFICULTIES

There were couple of difficulties that I faced during this internship. The main difficulty was requirement of a Nvidia GPU for deep learning. This was a main problem for me because my laptop’s GPU was AMD Radeon and I couldn’t use it for the BEVFusion inference. I also couldn’t do only CPU inference for BEVFusion because it would take a lot of time. My supervisor at the company recommended me to set up a ssh server to my home desktop which had an Nvidia GPU. I set up and ssh server and installed a software called Netbird in order to connect to the ssh server without sharing the same network. Then when I

tried to do inference on BEVFusion, I get an error which indicated that my GPU vram wasn't enough for BEVFusion inference. So, I tried to get a temporary computer with Nvidia GPU from the company but I couldn't get one due to company policy of not giving computers to short term interns and finding a specific computer with a Nvidia GPU which can run BEVFusion wasn't an easy job. So, I researched platforms with free online GPUs that I can use. I found that Google Cloud gives 300\$ of credit that I can use on GPU. I set up a virtual machine on Google Cloud and do all the inference steps once more on that virtual machine. I successfully completed BEVFusion Inference but I couldn't do the Accelerated BEVFusion Inference due to requirement of compute capability $> \text{sm_80}$ which the GPU that I used didn't pass the requirement. Also, I had problem with clashing libraries. In the README file at the code they only gave intervals for the library versions so I needed to figure out which versions are compatible which by searching and trying. Also, BEVFusion code is archived at Github and no longer is maintained. Because of the reason I needed to figure out how to fix errors I encountered at the code.

4.3 NEW SKILLS

I learned how to set ssh server in a computer and how to connect it remotely. I also learned how to create virtual machines in google cloud. While working with BEVFusion I learned about using Cuda, TensorRT, Nvidia Toolkit and I had an experience with Deep Learning and Vision Recognition. Moreover, I learned some new Linux commands.

5. CONCLUSIONS

During this internship, I was able to research various vision detection algorithms and learn how they work, what the difference is between them, and Why BEVFusion is more preferable than others for autonomous cars. I also had the chance for doing an inference with

BEVFusion. During the inference, I faced hardware problems and code errors. After fixing those problems, I managed to do segmentation and detection. After that, I used visualisation to visualise them. After finishing BEVFusion inference, I also tried to do Accelerated BEVFusion Inference, but I couldn't finish it due to hardware incompatibility. The remaining parts of the project, which are Accelerated BEVFusion Inference and setting BEVFusion on Nvidia Orin Drive can be finished by another intern later on. At the end of this internship, I got experience at how to do inference on a vision recognition algorithm, and I learned what CUDA, TensorRT, Nvidia Toolkit are. To sum up, even though the internship period was relatively short, it was a valuable experience for me to learn about vision detection and the autonomous vehicle industry.

6. RECOMMENDATIONS

I have two main recommendations for students who want to work at deep learning and vision recognition. First one is they should buy a computer with Nvidia GPU. By doing this, they will be able to do inference by using their GPU instead of renting remote GPUs or using their CPU. Second one is they should definitely take CS 415 (Introduction to deep learning), CS 419 (Digital Image and Video Analysis), EE 417 (Computer Vision). Those courses will definitely be useful in this field.

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